

Aparnaa Senthilnathan

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EDUCATION

Rochester Institute of Technology

B.S. and M.S. in Computer Science, RIT Presidential Scholar, GPA: 3.36

Rochester, NY

Aug 2021 - May 2026

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS

Full-Stack and Backend Engineering: React, FastAPI, REST APIs, Node.js, JavaFX, Streamlit, Git, Docker, Linux

AI, ML and Data Science: PyTorch, TensorFlow, scikit-learn, OpenCV, Librosa, Pandas, NumPy, Model Evaluation, Recommendation Systems

Creative AI and Interactive Systems: Three.js, WebGL, WebGPU, TouchDesigner, Unreal Engine, Ableton, Gemini API, LangChain

EXPERIENCE

Computer Vision Intern

Kitware, Inc.

May 2024 – August 2024

Carrboro, NC

- Orchestrated distributed machine learning data-processing workflows using Slurm and multi-GPU environments to support high-throughput model training, evaluation, and dataset-quality workflows
- Automated large-scale dataset curation and validation using FiftyOne and PyTorch, standardizing edge-case detection and improving data-quality checks

Artificial Intelligence Research Intern

Griffiss Institute

July 2023 - December 2023

Rome, NY

- Designed adversarial stress-testing frameworks to evaluate system reliability, targeting zero-shot classification failures and out-of-distribution data anomalies
- Implemented automated analysis pipelines using similarity scoring and evaluation metrics to quantify model robustness and identify classification failure patterns

PROJECTS

MAARA: Multi-Agent Automated Repair Assistant | Academic Team Project

April 2026 - May 2026

- Built a modular RAG-based multi-agent software system using Python, Ollama, ChromaDB/Qdrant, and structured feedback loops to improve code-generation quality and reduce failed outputs
- Designed a five-agent workflow - Planner, Evaluator, Coder, Critic, and Repair - to decompose tasks, validate implementation plans, generate code, run feedback loops, and patch failures
- Implemented a Sentinel strategy supervisor to evaluate pre-generation plans, driving a 13-point pass-performance increase over single-agent baselines on HumanEval

Mood-y: Voice-Based Mood Music Recommender | Academic Team Project

March 2026 - May 2026

- Developed a user-facing full-stack machine learning application using FastAPI and React, exposing REST APIs that converted voice inputs into personalized music recommendations
- Trained and evaluated speech-emotion classification models that extract 112-dimensional Librosa audio features, achieving 95.8% accuracy through parallel processing
- Engineered a recommendation and ranking engine using cosine similarity, weighted scoring, and mood-based filtering to rank 81,000+ tracks based on user emotion signals

FrequencyPrint: An Automated Deepfake Detection System | Capstone Project

January 2026 - May 2026

- Implemented an AI-driven audio forensics tool using Python, PyTorch, and Librosa to convert audio into spectrogram representations for visual pattern analysis and synthetic artifact detection
- Trained CNN-based PyTorch models to detect synthetic artifacts in AI-generated covers, achieving up to 61.5% classification accuracy
- Presented research findings and technical methodologies at the Imagine RIT festival and the Graduate Poster Showcase